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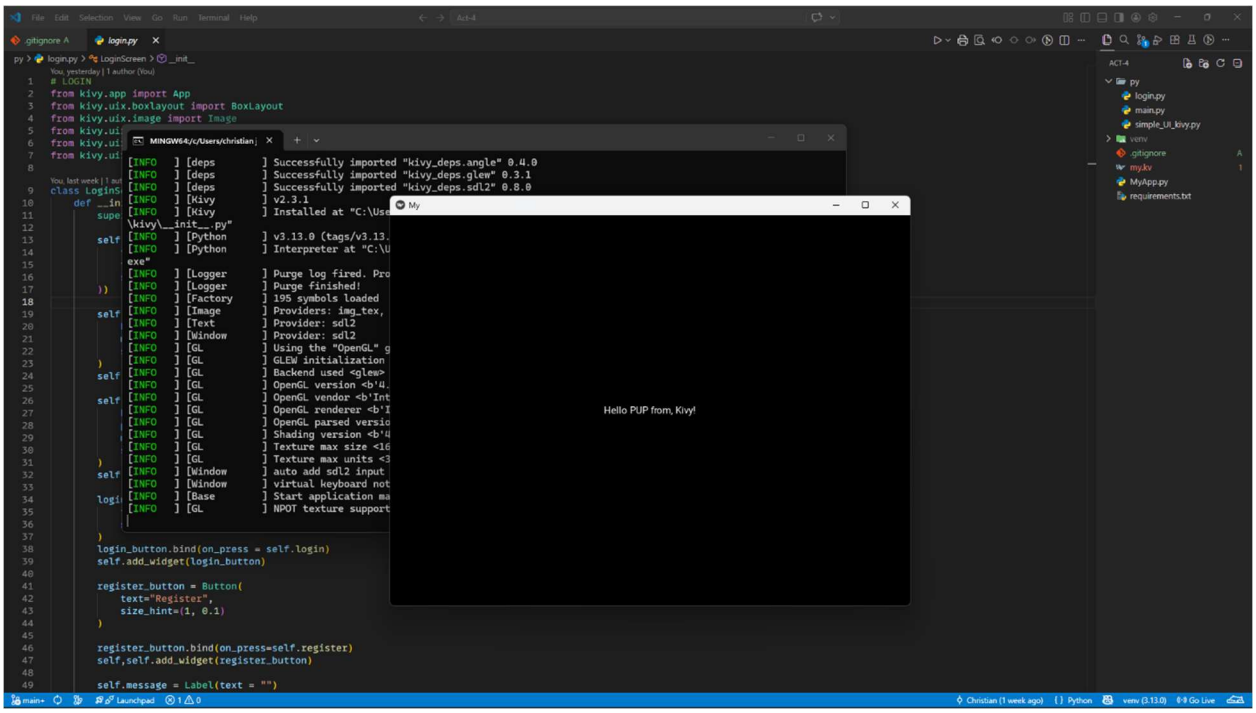
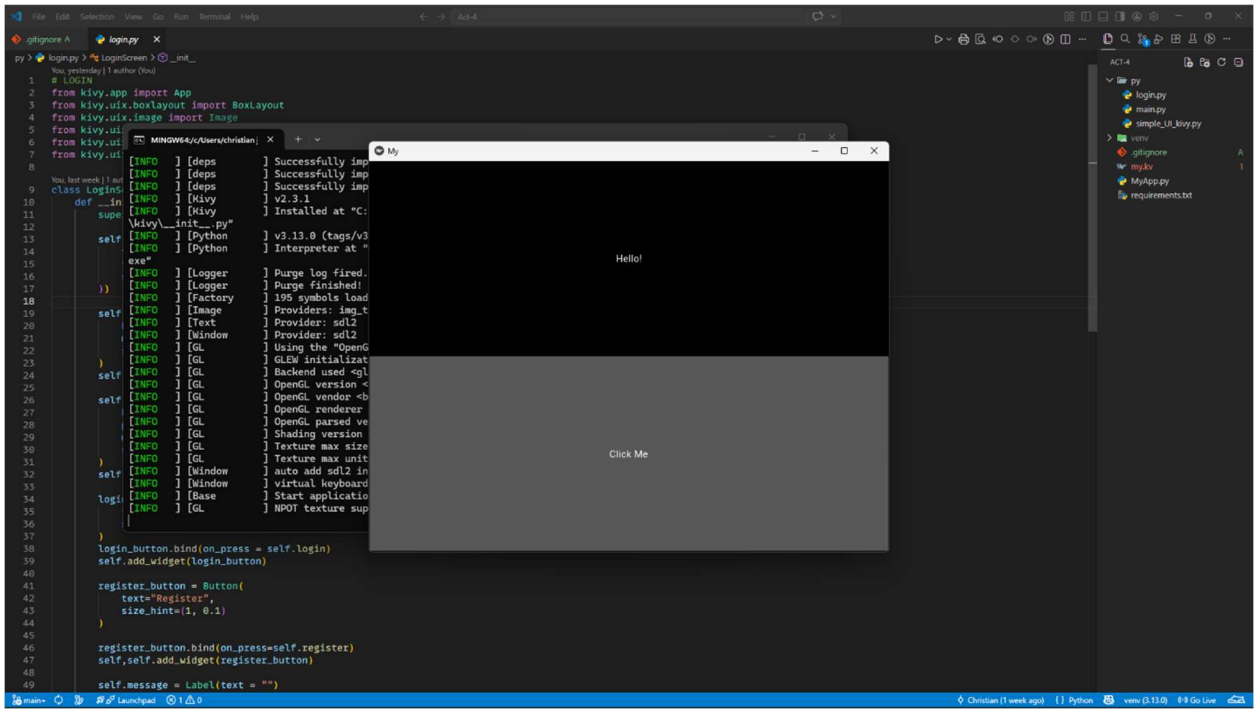
BSIT 3-2

ACTIVITY 4.1: KIVY SETUP AND BASICS

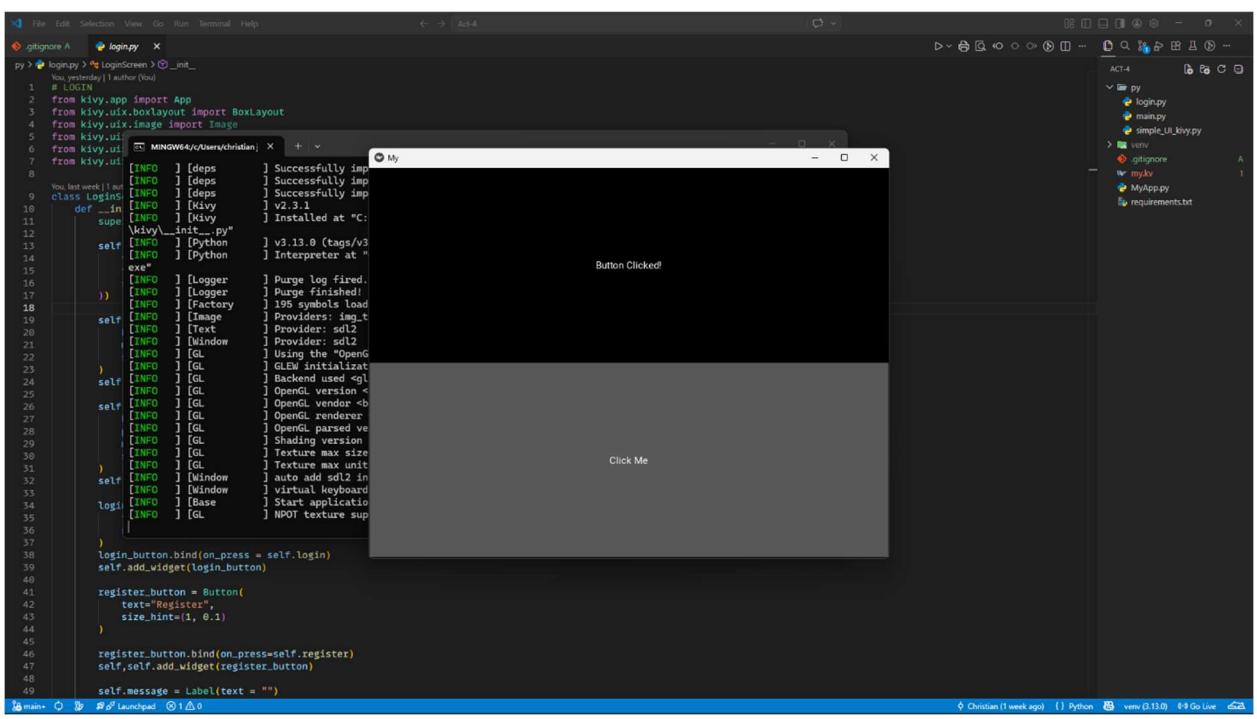
First Kivy App:



Simple UI:




Simple UI button is clicked:



Kivy vs Web Development

User interfaces can be developed using different tech stacks depending on the target platform. There are two common approaches when building user interfaces, Kivy and Web development. Kivy is a python-based framework for building cross-platform desktop and mobile applications.  Web development uses HTML, CSS and Javascript to build applications that run in a browser.

This table defines the difference between Kivy and Web Development.

| **Aspect** | **Kivy** | **Web Development** |
|---|---|---|
| Programming Language | Python | HTML, CSS, Javascript |
| UI Definition | Kv Language/Python | HTML |
| Styling | Kv Language/Python | CSS |
| Logic Handling | Python | Javascript |
| Runtime Environment | Native App Window | Web Browser |

Kivy runs as a standalone application that uses OpenGL for rendering UI components and does not require a web browser to show output. Web Development runs inside a browser; UI is rendered using the browser's rendering engine and requires a client-server model.

Layout and UI Structure are also handled differently. Kivy uses BoxLayout, GridLayout, FloatLayout, StackLayout to organize and design widgets. Web development uses Flexbox, CSS Grid, Positioning(Absolute, Relative, etc.). Kivy uses widget tree as UI hierarchy while Web development uses DOM or Document Object Model. In terms of responsiveness, Kivy uses pre-defined functions as size_hint, post_hint while Web Development have tools like CSS media queries, flex or grid.

Event Handling is also different from each two but have the same concept.

Kivy: `button.bind(on_press=my-function)`.

Or in Kv language: `on_press: root.my-function()`.

While Web Development have: `button.addEventListerner("click", myFunction);` using JS.

For Styling and Design: Kivy styling is done via KV or Python, while web apps use CSS and frameworks like Bootstrap or Tailwind.

For Navigation: Kivy uses ScreenManager, while web apps use URL routing.

I conclude that there are no superior or inferior to one another, Kivy is ideal for Python-based mobile/desktop apps, while web development is best for scalable, internet-based applications.